

Intelligent Association



Presented by

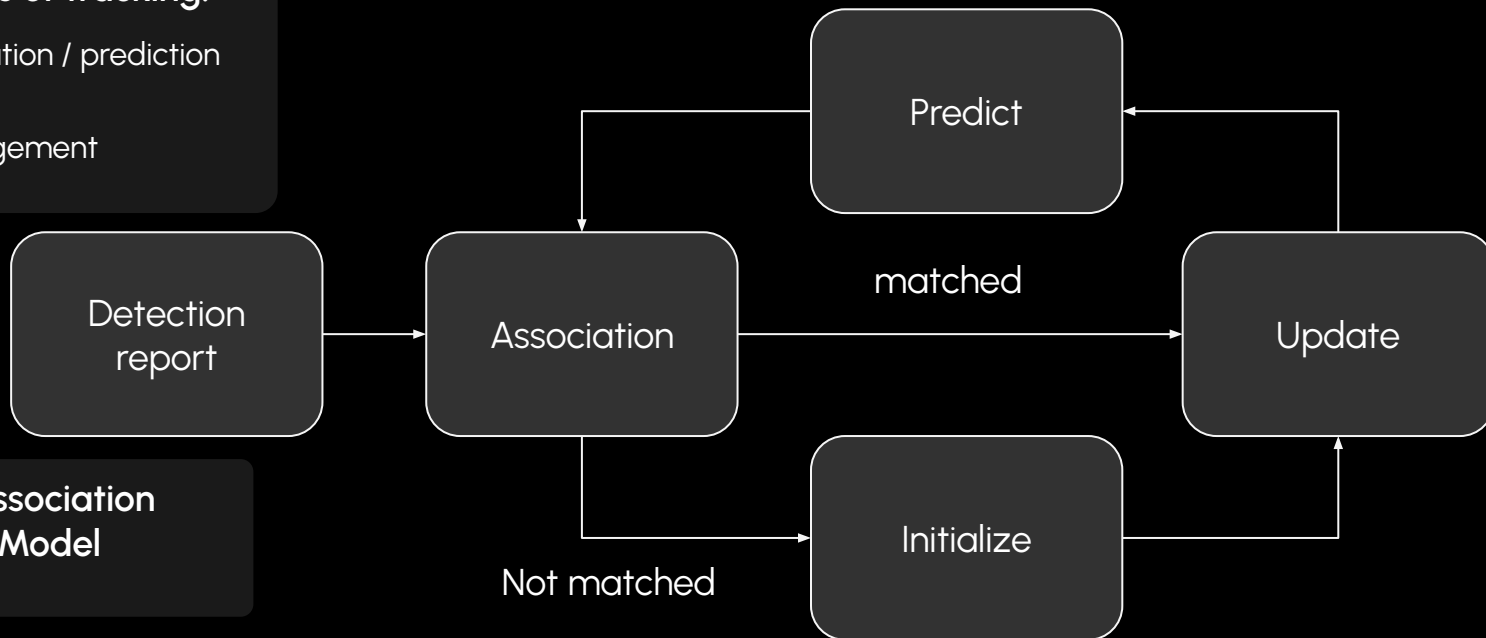
Spencer

Randolph

Background - Tracking

Three main parts of tracking:

- State estimation / prediction
- Association
- Track Management



GOAL: Improve Association Through Learned Model

Milestone 1:

Data Generation

Milestone 2:

Tracking Architecture
Model Training

Milestone 3:

Model Training
Evaluation

Planned work

Tracking / model:

further training and tuning

further state estimation exploration:

- adaptive kalman filter

Further association / feature set:

- Rcs likelihood model
- Motion explainability model

bug fixing

Evaluation:

- Compare to JPDA, PDA ?
- Test generalisability
- Model metrics ~ f1

Kalman filter updates

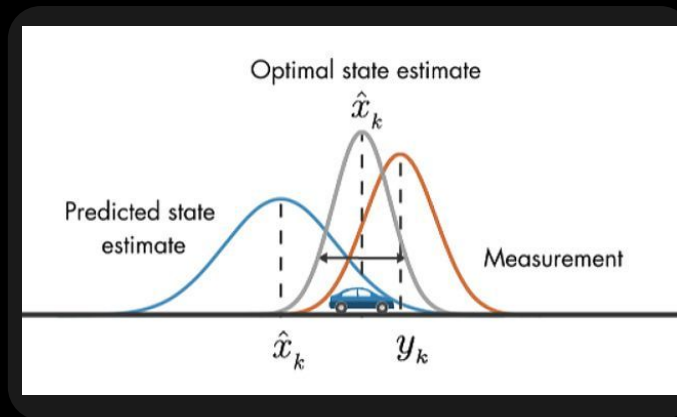
Problem:

- real world objects don't follow linear models
- process noise is limited by measurement noise

solution:

Adaptive kalman filter

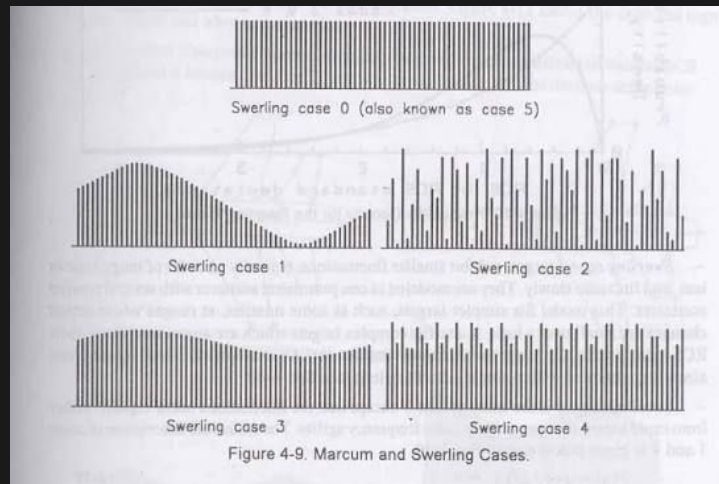
- Exponential moving average of residual
- Threshold of 2.25
- Triggers a 1.5 increase of process noise



Rcs likelihood

Intuition:

- RCS's can be represented by swerling types
- Estimate the rcs of a target through the radar range equation



Implementation:

- μ : rcs mean
- σ : rcs std
- X : incoming estimate

$$\text{Cost} = |x - \mu| / \sigma$$

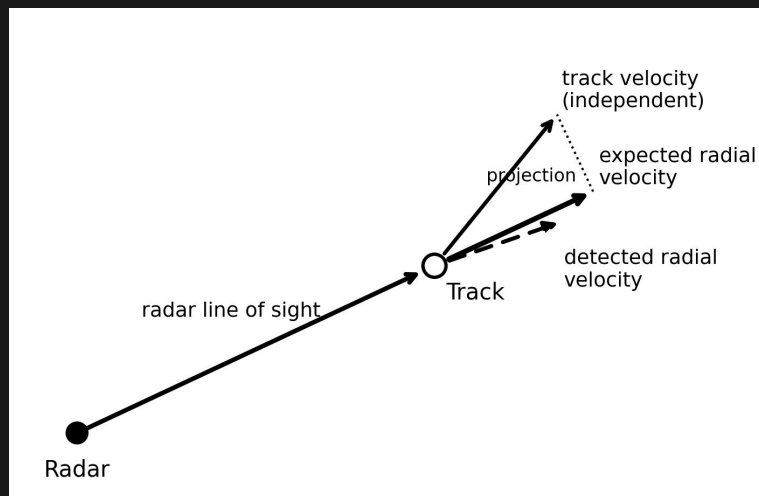
Model the difference between type 1,2 and 3,4

Require increase track age to be accurate

Radial consistency

Intuition:

- Isolates the motion of the radar from position
- Improve association of close targets



Implementation:

- Assumes radar is (0,0)
- Constant Variance

$$v_{r,pred} = \frac{xv_x + yv_y}{\sqrt{x^2 + y^2}}$$

$$d_{v_r} = \frac{|v_{r, det} - v_{r, pred}|}{\sigma_{v_r}}$$

Model training

2 models
70 tracking runs
518,000 pairs

Input:
4 association costs
Track age
Miss count

Data set:
Incorrect 483,794 (93.33%)
Correct: 34,588 (6.67%)

Standard KF model:
Accuracy: 98.18%
Precision correct: 85.21%
F1: 86.78%

Adaptive KF model:
Accuracy: 98.20%
Precision correct: 85.35%
F1: 86.87%

Kalman filter comparison

- Idealized world vs real world

$$\text{MOTA} = 1 - \frac{FN + FP + IDSW}{GT}$$

Dataset	Correct trigger %	Not correct trigger %
Easy	88.52%	11.48%
Medium	41.18%	58.82%
Hard	14.67%	85.33%
Non-similar	21.21%	78.79%

Dataset	Learned + Standard KF	Learned + Adaptive KF
Easy	0.797	0.731
Medium	0.436	0.206
Hard	-0.095	-0.518
Non-Similar	0.059	0.033

Tracking base lines

GNN

- 1 to 1 tracker
- Global assignment to chose closest track

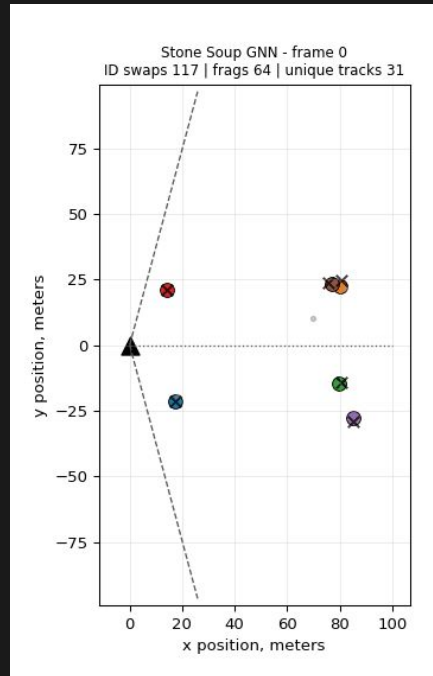
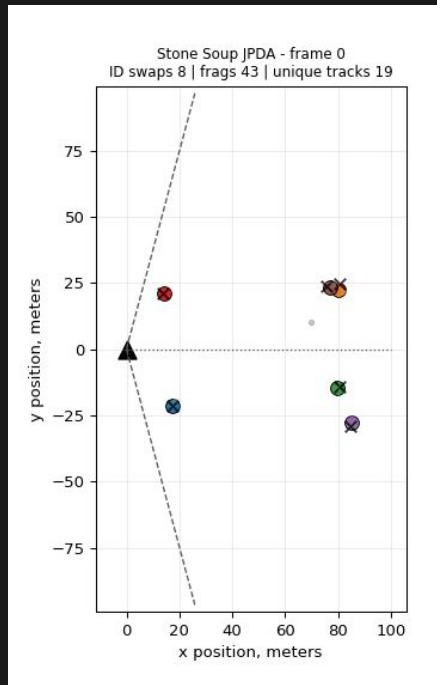
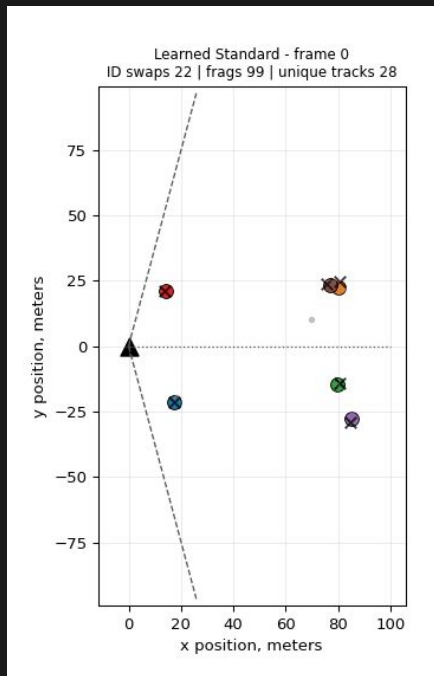
JPDA

- Associates the probability of a track being assigned to all feasible detections
- Updates tracks based of off those weights

Tuning

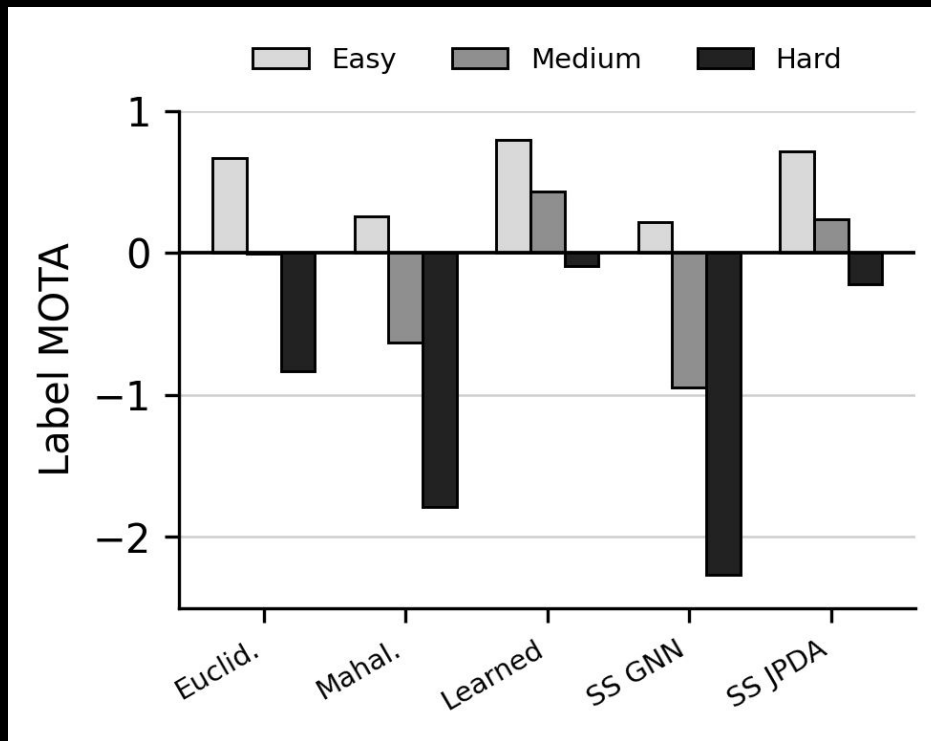
- Fair comparison
- Tested range of gates ... on small sample size

Results Medium Data Set



Results

- Similar mota for learned and JPDA
- Learned associates more often but has more incorrect
- Jpda more conservative and struggles with near target identity



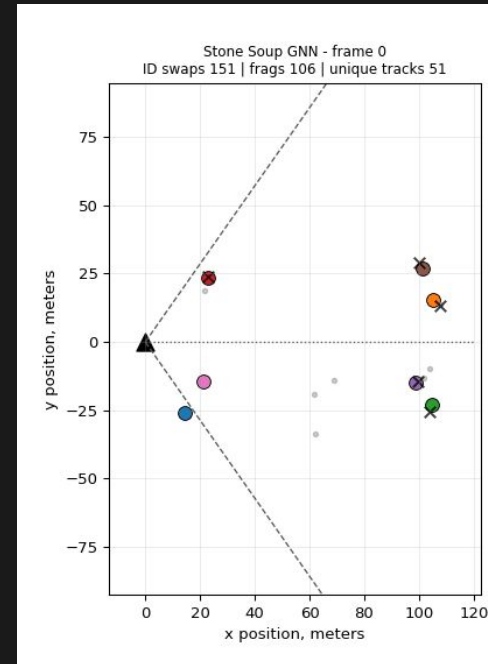
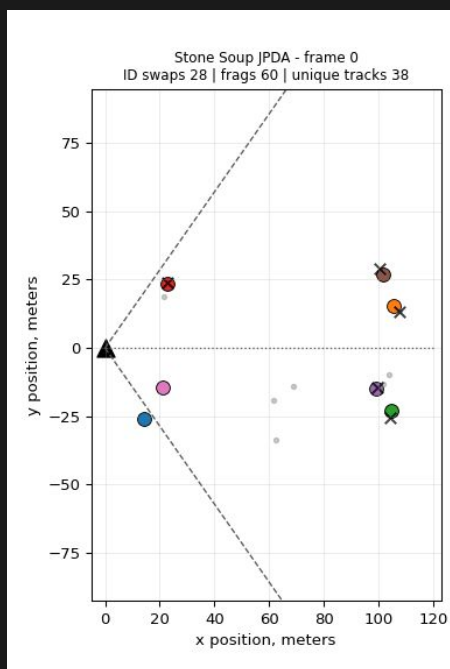
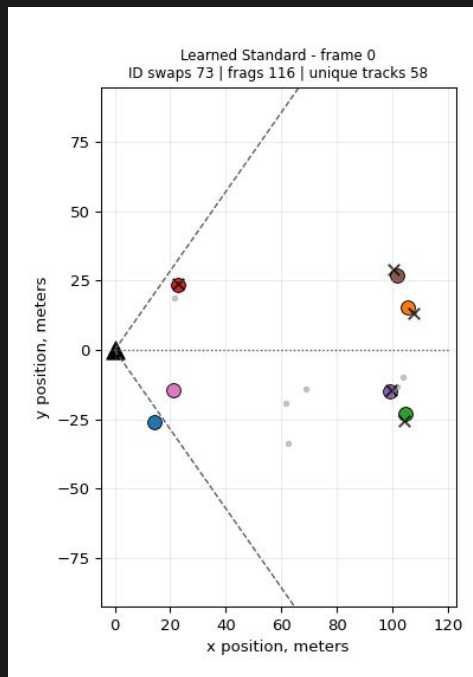
Generalizability

See how the algorithm holds up when detection level data shifts

- Change fov ~ different pos
 - Fov 75 -> 50
- Increased radial velocity sampling
 - 10 m/s -> 15 m/s
- Shifted RCS
 - Mean 12 -> 8

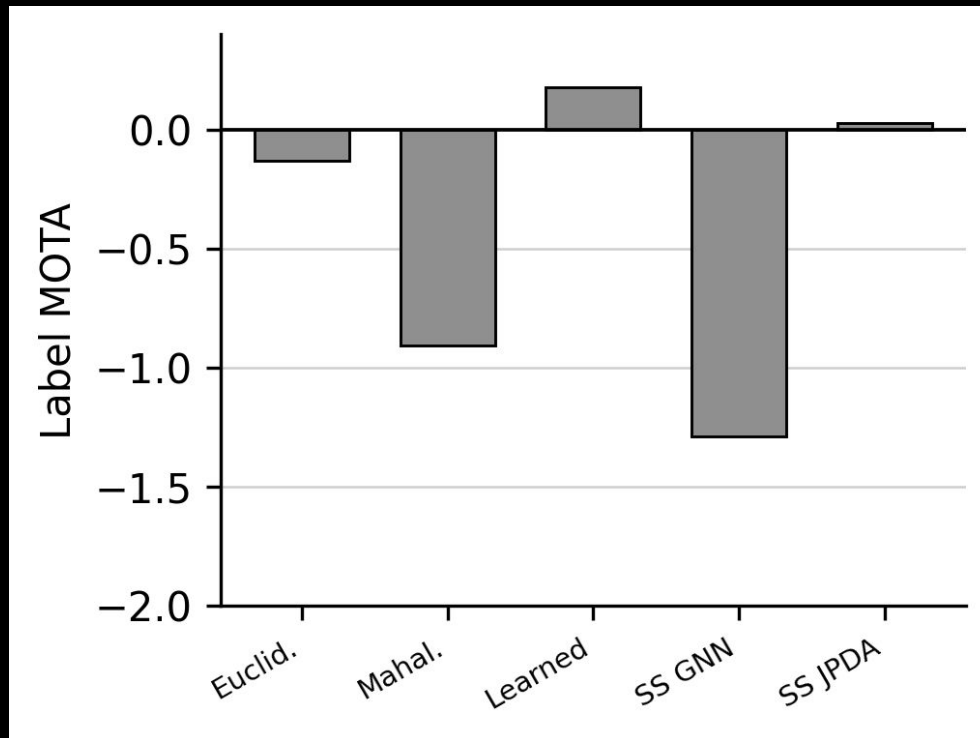
Further generalizability for milestone 3+

Non-Similar Results



Results

- Generalizability holds similar across learned and jpda filter



Questions?